

10 Sept 24

23 BC AND 410

Assignment - 2Software Engineering

SECTION - A

Ques 1 What are functional ~~require~~ requirements in Software engineering?

Ans 1 functional requirements specify what a System should do. They define the System's functionality or behavior in respect response to specific inputs or conditions. functional requirements describe tasks, functions, and behavior in response to specific inputs or conditions.

Ques 2 Define non-functional requirements and provide an example.

Ans 2 Non functional requirements define how a System should perform, focusing on quality attributes like performance, reliability, scalability, and usability.

Example → The system must handle upto 10000 concurrent users without performance degradation is a non functional requirement focused on scalability.

Ques What is the purpose of a data flow ~~modeling~~ diagram (DFD)?

Ans Purpose of data flow diagram is to visualise the flow of data within a system. It represents how data moves between processes, data stores, and external entities. The main purpose of DFD is to help stakeholders understand how information is processed and where it flows, making it easier to analyse the system's functionality.

Ques Explain the importance of use case modeling.

Ans Use case modeling is important because it captures user requirements, facilitates communication between stakeholders, improves system design, provides a basis for testing, helps define scope, and ensures user centric development.

Ques What is the role of interface specification in the system requirements?

Ans The role of interface specification in the system requirements is to define how different system components or external system interact. It details the inputs, outputs, data formats, protocols, and communication methods ~~are~~ necessary for seamless integration, ensuring compatibility, reducing errors, and enabling smooth data exchange between system parts or with external entities.

SECTION-B

Ques Differentiate between functional and non functional requirements with example

Aspects	Functional Requirement	Non functional Requirement
Definition	Describe what the system should do.	Describe how the system should perform and behave.
Focus	Specific functionalities and interactions.	Quality attributes and system performance.

Purpose	Specify actions, tasks and processes the system must perform.	Define constraints and performance characteristics.
Nature	Operational task and features.	Performance metrics and quality standards.
Examples	User can login with username and password. System can generate reports.	System must support 1000 concurrent users. System must be available 99.9% of the time.

Ques 2 What is the significance of creating entity relationship diagram during system analysis?

Ans 2 Entity Relationship ~~mod~~ Diagram (ERD's) play a crucial role during system analysis by providing a visual representation of the system's data structure and its relationships. The significance of creating ERD's includes the following:

1. Clear Data representation → Visualizes data structure and flow.
2. Facilitates communication →

- Helps stakeholders understand the system.
- Improves System Design → Ensures efficient and normalized ~~xxxxxx~~ database design.
- Identifies Relationships → Highlights entity relationships and dependencies.
- Supports Database Creation → Acts as a blueprint for database schema design.
- Aids error detection → Identifies potential design flaws early.
- Enhances flexibility → Simplifies future system changes and scalability planning.

Ques 3 Explain the purpose and use of content diagrams in system modeling.

Ans 3 Purpose of Content Diagram →
 1. Define System Boundaries → Shows what is inside and outside the system.

- ② Highlight External Interactions → Illustrates connections with ~~the~~ external entities (users, systems).
- ③ Simplify Complex Systems → Provides a simplified, high level view of system.

Use of Context Diagram →

- ① Initial phase of design → Used early to visualize system boundaries and interactions.
- ② Stakeholder Communication → Helps explain the system's scope and environment to stakeholders.
- ③ Requirements Gathering → Aids in identifying inputs, outputs and external influences during requirement analysis.

SECTION-C

Ques 1

Discuss the various stages of requirements engineering. How do functional and nonfunctional requirements impact the software development process?

Ans 1

Stages of requirement engineering are →

- ① Requirement Elicitation → Gathering requirements from stakeholders using interviews, surveys, etc.
- ② Requirement Analysis → Analyzing, categorizing and resolving conflicts in the gathered requirements.
- ③ Requirement Specifications → Documenting the analyzed requirements in a structured format (eg → SRS).
- ④ Requirement Validation → Ensuring the requirements are accurate, complete and feasible through reviews or prototypes.
- ⑤ Requirements Management → Managing changes and maintainability, traceability throughout the project lifecycle.

Impacts →

functional Requirements →

- ① Define what the system must do (features like login, data processing)
- ② Directly impact system design and implementation
- ③ Missing or incorrect functional requirements can lead to system failures or redesigns.

Non-functional Requirements →

- ① Define how system performs (performance, security, usability).
- ② Influence architecture and technology choices.
- ③ Ignoring them can lead to poor performance, security issues, and bad user experience.

Ques 2 Explain the concept of use case modeling in detail. How does it assist in object oriented modeling?

Ans 2

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Use Case Modeling →

- Use case modeling captures system functionality from a user's perspective, focusing on what the system should do to meet user goals.

Key Elements →

- ① Actors → External Entities interacting with the system (users or other systems).
- ② Use Cases → Specific interactions between actors and the system to achieve goals.
- ③ System Boundary → Defines the scope of the system.
- ④ Associations → Relationships between actors and use cases.

Assisting Object Oriented Modeling →

- ① Identifies object/classes → Uses cases help discover key objects and classes.
- ② Defines object responsibilities → Shows what actions object should perform.
- ③ Clarifies object interactions → Helps model relationship and interactions between objects.

- ① Derives Class/Sequence Diagrams → Translates use case flows into object ~~interaction~~ interaction for detailed design.

Ques (Compare and contrast the roles of data flow diagrams (DFD) and entity relationship diagrams ~~and~~ (ERD) in software analysis. How do these tools support process modeling?

Aspects	Data Flow Diagram	Entity Relationship Diagram
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Purpose	Models the flow of data through a system processes.	Models the relationships between entities (data objects) in a system.
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Focus	Focuses on the process (what the system does) and how data flows between these processes.	Focuses on the data itself (entities) and their relationships.
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Components	Processes, data stores, data flows, external entities.	Entities, attributes, relationships, cardinality.
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Usage	Data Flow Diagram	Entity Relationship Diagram
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Usage	Used in process modeling to define how data moves in the system.	Used in database design to model data relationships.
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Diagram levels	Often hierarchical with multiple levels	Typically one detailed diagram shows showing entities and their relationships.
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Examples	Depicting how customer orders flow through an e-commerce e-commerce system	Showing how customers, orders and products relate in a database
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